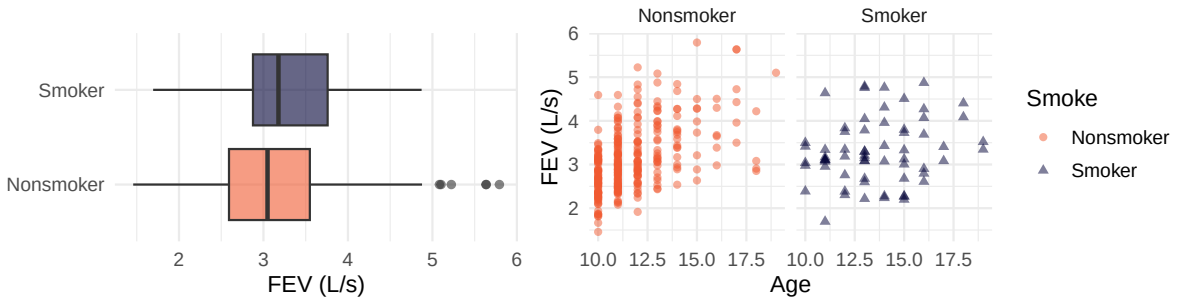
# Day 10: Adding Categorical Predictors

## New data: lung function and smoking

**Goal:** investigate the association between smoking and lung capacity using data from 345 adolescents ages 10-19.

| Variable | Description |
|---|---|
| FEV | forced expiratory volume (liters/second) |
| Age | age in years |
| Smoke | Smoker or Nonsmoker |

## EDA




Notice smokers skew *older* – any raw FEV difference between smokers and nonsmokers is tangled up with age. This is exactly why we need a model that controls for both variables at once.

## Indicator variables

Regression requires a **numeric** representation of every variable, so a categorical predictor gets converted to indicator (0/1) columns (e.g. `Smoker` = _____, `Nonsmoker` = _____)

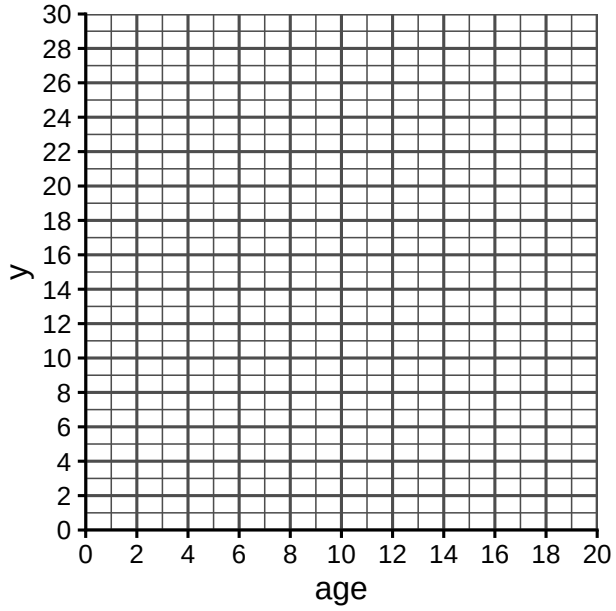
# Group Task 1

For each regression model, sketch the fitted model. To do this, first write the expression for the fitted model for non-smokers and the fitted model for smokers (i.e., plug in 0 and 1 for the `smoker` indicator variable). Once you have these two equations, use them to sketch the lines on the provided graph. (*Note:* fake equation alert! not estimates from the real data)

**Model 1:** $\hat{\mu}(y|x) = 10 + 1\texttt{age} - 2\texttt{smoker}$

Fitted equation for non-smokers:

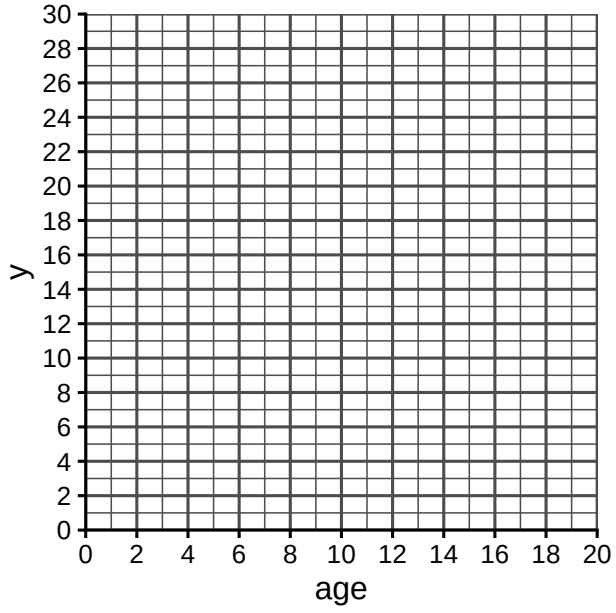
Fitted equation for smokers:




**Model 2:** $\hat{\mu}(y|x) = 5 + 1\texttt{age} - 0.5\texttt{age} \times \texttt{smoker}$

Fitted equation for non-smokers:

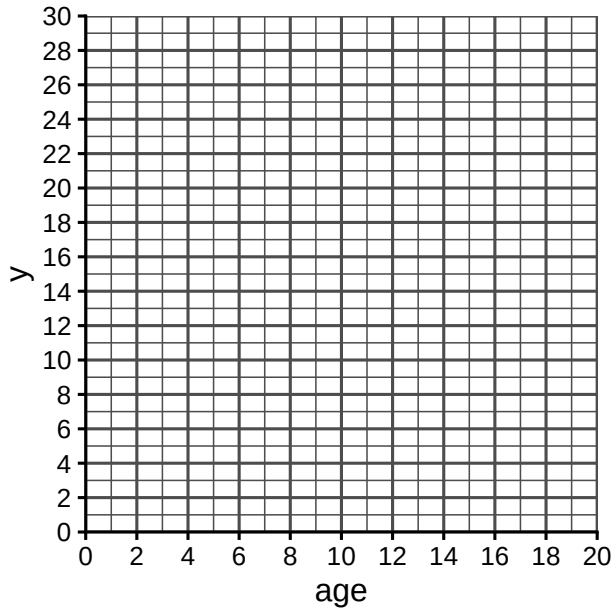
Fitted equation for smokers:



**Model 3:** $\hat{\mu}(y|x) = 4 + 0.5\texttt{age} + 3\texttt{smoker} - 0.5\texttt{age} \times \texttt{smoker}$

Fitted equation for non-smokers:

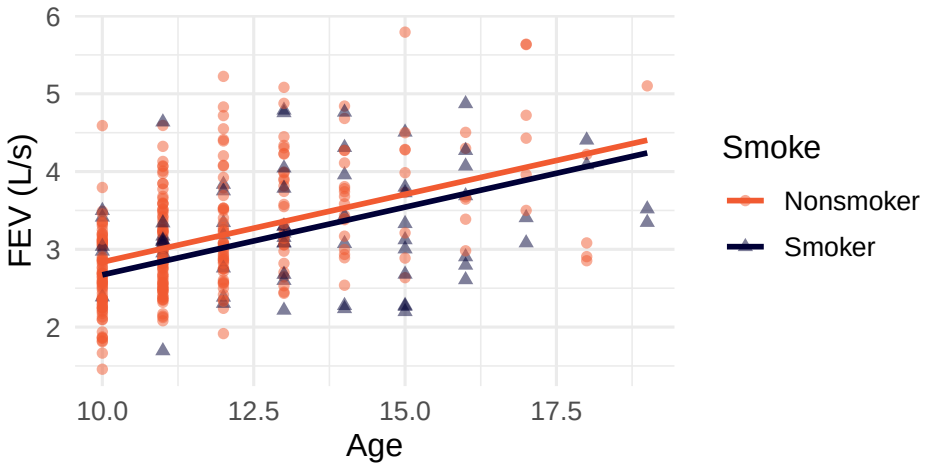
Fitted equation for smokers:




> **!** Stop here
>
> Let Amanda know when your group finishes!

## Three ways to combine a line + a group

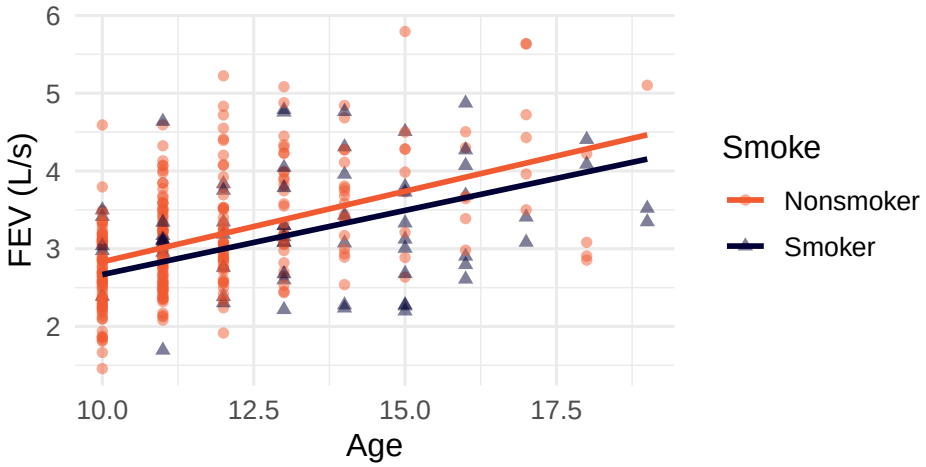
1. _____ model: Same rate of increase in both groups, but a different baseline.



$$\mu(y|x) = \beta_0 + \beta_1 \texttt{age} + \beta_2 \texttt{smoker}$$

- $\beta_0$: intercept for nonsmokers
- $\beta_0 + \beta_2$: intercept for smokers
- $\beta_1$: slope for _____

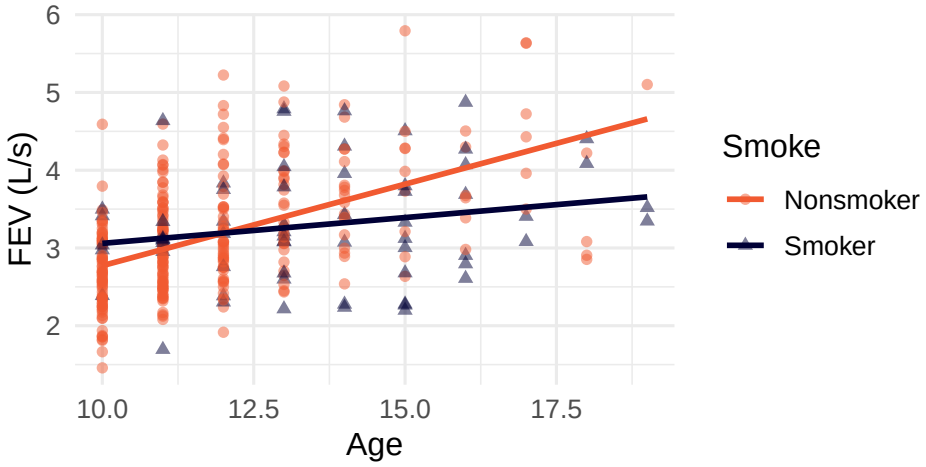
2. _____ model: Same baseline (at age 0), but age affects FEV differently for smokers and nonsmokers.




$$\mu(y|x) = \beta_0 + \beta_1 \texttt{age} + \beta_2 \texttt{age} \times \texttt{smoker}$$

- $\beta_0$: intercept for _____
- $\beta_1$: slope for nonsmokers
- $\beta_1 + \beta_2$: slope for smokers

3. _____ model: Both the baseline *and* the rate of increase differ by group.



$$\mu(y|x) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{smoker} + \beta_3 \text{age} \times \text{smoker}$$

- $\beta_0$: intercept for nonsmokers
- $\beta_0 + \beta_2$: intercept for smokers
- $\beta_1$: slope for nonsmokers
- $\beta_1 + \beta_3$: slope for _____

Fitting the separate lines model in R:

```
fev_sep <- lm(FEV ~ Age * Smoke, data = fev)
coef(fev_sep)
```

```
    (Intercept)             Age     SmokeSmoker Age:SmokeSmoker
      0.6747724       0.2097247       1.7197133      -0.1432927
```

R names an indicator after the level that is coded **1**. The level that never appears in a coefficient name is the reference level.

Here, the reference level is _____.

| R output | Model term | Meaning |
|---|---|---|
| `(Intercept)` | $\beta_0$ | intercept for nonsmokers (reference level) |
| `Age` | $\beta_1$ | slope for nonsmokers |
| `SmokeSmoker` | $\beta_2$ | **difference** in intercepts (smokers — nonsmokers) |
| `Age:SmokeSmoker` | $\beta_3$ | **difference** in slopes (smokers — nonsmokers) |

**Example:** Write the fitted regression line for nonsmokers and for smokers.

**Example:** A TRUE/FALSE predictor

`weekend` in the `bikes` data is TRUE/FALSE, so R treats TRUE as 1 and names the coefficients `weekendTRUE` and `temp_actual:weekendTRUE`:

```
coef(lm(rides ~ temp_actual * weekend, data = bikes))
```

```
          (Intercept)           temp_actual            weekendTRUE
        -1843.8674329            87.1729679          -714.9382143
temp_actual:weekendTRUE
            0.4110155
```

## More than two categories

**Motivating example:** Let's say we want to incorporate `season` (fall, spring, summer, winter) from the `bikes` data.

**A bad idea:** convert the column to a single number: Fall $\to 1$, Winter $\to 2$, Spring $\to 3$, Summer $\to 4$, then fit $\mu(\text{rides} \mid \text{season}) = \beta_0 + \beta_1 \text{season}$.

This is wrong. It forces the seasons onto an arbitrary numeric scale and assumes that the "jump" from fall to winter is the same size and direction as the jump from spring to summer.

**A better idea**: indicator (dummy) variables

For a categorical variable with $k$ levels, create $k-1$ indicator variables (one level is left out as the **reference level**).

| season | seasonwinter | seasonspring | seasonsummer |
|---|---|---|---|
| winter | 1 | 0 | 0 |
| fall | 0 | 0 | 0 |
| spring | 0 | 1 | 0 |
| spring | 0 | 1 | 0 |
| fall | 0 | 0 | 0 |
| summer | 0 | 0 | 1 |

The reference level in this case is _____.

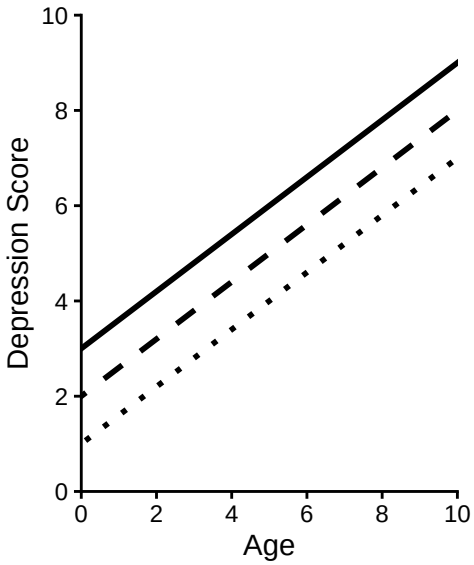
## Group Task 2

Researchers investigated the association of depression with age and education, based on a nationwide (U.S.) telephone survey of 2,031 adults aged 18 to 90. Of particular interest was their finding that the association of depression with education strengthens with increasing age—a phenomenon they called the "divergence hypothesis."

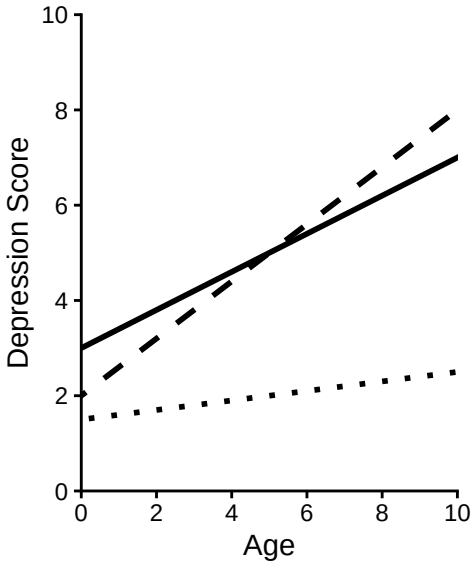
The researchers constructed a depression score from responses to several related questions. Education was categorized as (i) college degree, (ii) high school degree plus some college, or (iii) high school degree only. In the sketches below, the line type represents education: solid = college degree, dashed = high school plus some college, dotted = high school only.

1. Define indicator variables that represent the `education` category. (Start by determining how many indicators you need. Then define them.)

2. Suppose that one researcher wants to fit a model that represents the below sketch. Carefully write the mean function for the regression model that you should fit.




3. Suppose that one researcher wants to fit a model that represents the below sketch. Carefully write the mean function for the regression model that you should fit.




**!** Stop here

Let Amanda know when your group finishes!